TGL Lagrangian and operator structure

$$\mathcal{L}_{\mathrm{TGL}} = \mathcal{L}_{\mathrm{LCDM}} \cdot (1 + \beta_{\mathrm{TGL}} \cdot |1 + w_{\mathrm{eff}}|)$$

Hilbert decomposition

$$\mathcal{H} = \mathcal{H}_{2D} \oplus \mathcal{H}_{Q}$$

2D: Name + Verb
(boundary sector)

$$P_{2D} + Q = \mathbb{I}$$

Q: Word
(inert charge carrier)

Unique GKSL operator: $L_k = \sqrt{\beta_{\mathrm{TGL}}} \cdot \sqrt{K_\partial}$